

Passive Piezo Sounder

Audible cues without blocking the loop

Own the timer, pass time explicitly, and make silence the safe state.

Lesson	005 — component	Time	80–110 minutes
Requires	Digital output and explicit time	Board	Arduino Mega 2560
Output	Passive piezo on D6	Status	host verified; hardware experimental
Safety	E1: identified logic-level passive piezo only	Power	Mega USB only

1 Why sound is a resource problem

A tone looks like a simple output, but it couples a pin to a hardware timer. ADK’s `PiezoSounder` claims both resources before it touches the circuit. If either is busy, initialization fails predictably. Once initialized, `play()` starts a square wave and returns immediately. The main loop remains free to scan buttons, update LEDs, and enforce deadlines.

Learning targets

By the end, you can:

- distinguish a passive piezo transducer from an active buzzer or speaker;
- wire a modest passive piezo load to Mega D6 with a series resistor;
- explain why tone owns D6 and Timer2;
- schedule and stop a tone using explicit `TimePoint` values;
- predict pin and timer claim conflicts before hardware changes;
- verify duration, replacement, shutdown, and rollover deterministically;
- show that stop, shutdown, and destruction return the endpoint to silence.

2 Part boundary and safety

Safety checkpoint. Use only a small *passive piezo transducer* documented for logic-level excitation. Do not substitute an 8Ω speaker, magnetic sounder, siren, active buzzer module, amplifier, or unknown part. Disconnect USB and every other power source before changing wiring. Begin with short tones at a comfortable distance; stop if sound is painful, startling, or causes discomfort.

Identify the part before building:

- A passive piezo needs an alternating waveform; steady DC produces at most a click. Kits may label it “passive buzzer”.
- An active buzzer contains an oscillator and sounds from steady DC. It is a different component and is not used here.
- A dynamic speaker is a low-impedance inductive load. A Mega pin must not drive it directly.

Transistor/speaker boundary. A speaker, high-current sounder, or louder transducer requires a separately designed driver: a suitable transistor or amplifier, current limiting, supply analysis, common-ground or isolation decision, and inductive protection where the load requires it. That driver is outside this lesson. A transistor is not permission to guess a load’s ratings.

Required: Mega 2560, USB data cable, breadboard, one passive piezo transducer, one 100Ω series resistor, one measured $10\text{ k}\Omega$ weak-pull resistor, a high-impedance oscilloscope or logic analyzer, a DMM, and jumper wires. Use the transducer datasheet when its permitted voltage or wiring differs. Record the manufacturer's part number, rated voltage, and capacitance or drive limits before energizing it. If the kit part cannot be identified well enough to establish those limits, keep the exercise host-only.

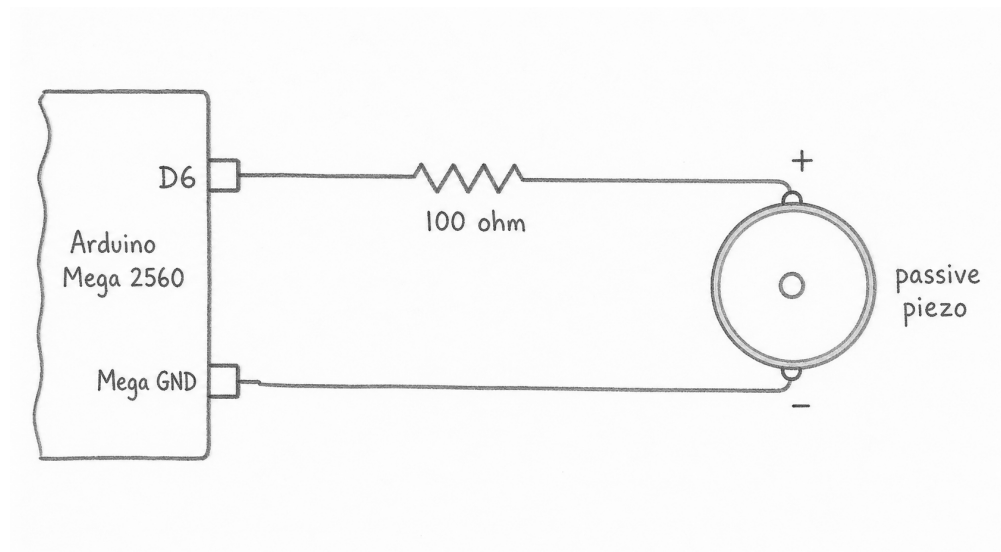
3 Predict before wiring

Action	Expected software state	Expected circuit state
Construct only	not initialized, not sounding	D6 untouched
Initialize	initialized, not sounding	D6 passive; Timer2 owned
Acquisition evidence	both components owned	D13 on 250 ms, then off 750 ms
Play 440 Hz for 250 ms	sounding at 440 Hz	square wave drives piezo
Update before deadline	still sounding	waveform continues
Update at/after deadline	not sounding	tone stopped; D6 restored to input
Shutdown or destruction	not initialized, not sounding	silence; claims released

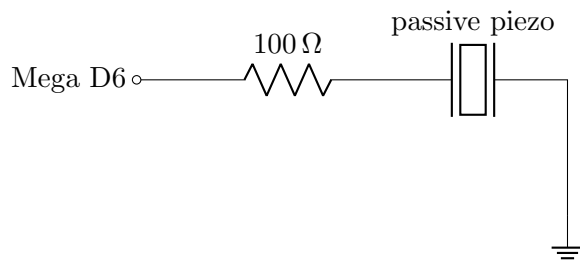
Conflict prediction: If another component owns Timer2, which status should `initialize()` return?

Should D6 change mode? _____ Why?

4 Exact Mega 2560 circuit



Orientation only. This drawing identifies the intended parts and signal path, but pin spacing and board geometry are illustrative. The exact schematic and connection table below are authoritative. Trace D6, 100Ω , passive piezo, and Mega GND before applying power.



Use one Mega GND pin.
If the piezo is marked +,
place + toward D6.

From	Through	To
Mega D6 piezo return / – terminal	100 Ω resistor jumper	passive piezo signal / + terminal Mega GND

The resistor limits edge current into the piezo’s capacitance and makes the exercise conservative; it does not make an arbitrary load safe. Do not connect a separate 5 V wire. Keep leads short. With power disconnected, verify no continuity from 5 V to GND and trace D6 through the resistor and piezo to GND.

5 Use the exact interface

```
#include <Adk.h>

constexpr adk::PinId          SounderPin = 6;
constexpr adk::PiezoSounder::Frequency CueHz      = 440;
const adk::Duration           CueLength  (250);

adk::Runtime      runtime;
adk::PiezoSounder sounder (runtime.resources (), SounderPin);

void setup ()
{
    const adk::Status status = sounder.initialize ();

    if (!status.ok ())
    {
        return;
    }

    const adk::TimePoint now (millis ());
    sounder.play (CueHz, CueLength, now);
}

void loop ()
{
    const adk::TimePoint now (millis ());
    sounder.update (now);
}
```

`play(frequency, duration, now)` accepts 31–20,000 Hz and a duration of 1–60,000 ms. Zero duration, an out-of-range frequency, or an excessive duration returns a status whose `error()` is `StatusCode::InvalidArgument` without starting a new tone. `play()` before initialization reports `error()` as `StatusCode::NotInitialized`.

The API does not call `delay()`. It stores the supplied start time and duration. Each `update(now)` compares the caller’s time with the deadline. The caller therefore controls replay, test time, and loop cadence.

5.1 Lifecycle and replacement

Construction only stores the registry and pin. Successful initialization claims D6 first, then Timer2. If Timer2 is unavailable, the D6 claim is rolled back. Repeated successful initialization is harmless.

A successful `play()` while already sounding replaces the frequency, duration, and start time. It does not queue a tone. `stop()` is idempotent: it calls the platform stop operation only while sounding, clears the audible state, and returns D6 to input mode. `shutdown()` stops first, releases Timer2 and D6, and is also idempotent. The destructor calls shutdown, so cleanup remains correct in an exception-using host environment even though ADK throws no exceptions internally.

6 Timer2 ownership

On the Mega 2560 target, this implementation models Arduino tone generation as exclusive ownership of timer index 2. Timer2 is also associated with hardware PWM channels on Mega pins 9 and 10. Board-aware PWM code and PiezoSounder must therefore agree through the same `ResourceRegistry`; incompatible use fails instead of silently changing frequency or breaking a tone.

Existing owner	Sounder initialization	Required evidence
D6 pin owner	<code>error(): ResourceBusy</code>	no pin mode or tone call
Timer2 owner	<code>error(): ResourceBusy</code>	D6 claim rolled back; no hardware call
Unrelated pin/timer owner	<code>ok(): true</code>	D6 and Timer2 claimed
Same initialized sounder	<code>ok(): true</code>	no repeated hardware work

Safety checkpoint. A registry can prevent conflicts only when every participating component uses that registry and the board map describes the shared timer correctly. Direct calls to Arduino `tone()`, `analogWrite()`, or third-party timer libraries bypass ADK ownership. Do not mix them in this lesson.

7 Deterministic host tests

The implementation is host verified with fake Arduino tone operations. Add or review tests that drive exact time values; never sleep in a correctness test.

```
fake::reset ();
adk::ResourceRegistry resources;
adk::PiezoSounder    sounder (resources, 6);

require (sounder.initialize ().ok ());
require (sounder.play
  (
    440,
    adk::Duration  (250),
    adk::TimePoint (1000)
  ).ok ());

sounder.update (adk::TimePoint (1249));
require      (sounder.sounding ());
```

```

sounder.update (adk::TimePoint (1250));
require        (!sounder.sounding ());
require        (sounder.frequency () == 0);

```

Required deterministic cases:

1. construction makes no Arduino call;
2. initialization claims pin and Timer2 before touching hardware;
3. pin conflict and timer conflict return **ResourceBusy**;
4. timer-claim failure releases the temporary pin claim;
5. play before initialization returns **NotInitialized**;
6. boundary frequencies 31 and 20,000 Hz succeed;
7. frequency 30 or 20,001 Hz, zero duration, and 60,001 ms fail;
8. update one millisecond before the deadline preserves the tone;
9. update exactly at and after the deadline stops it;
10. a second play replaces the first tone and restarts its deadline;
11. explicit **stop()**, repeated stop, shutdown, repeated shutdown, and destruction produce the minimal safe trace;
12. elapsed-time arithmetic works across 32-bit millisecond rollover.

Rollover vector. Start at `TimePoint(0xffffffff0)`, play for 32 ms, update at `TimePoint(0x0000000f)` and then `TimePoint(0x00000010)`. The first update is one millisecond early; the second reaches the deadline.

8 Hardware investigation

The canonical sketch is finite and gives acquisition its own evidence. After both components initialize, TP-ACQ/D13 is lit for 250 ms and dark for 750 ms while the sounder remains silent. It then plays one 440 Hz, 250 ms cue and calls **shutdown()** on the D13 LED and sounder. It makes no further hardware calls. Pressing reset repeats the entire sequence. Thus the D13 pulse proves acquisition separately from the cue at TP-SND/D6 and from D6's released safe state.

1. With USB removed, identify the exact board, passive piezo, resistor, instruments, and probe input impedance. Label the D6 pin-side end of the 100 Ω resistor **TP-SND/D6**; label the factory LED signal **TP-ACQ/D13**. Trace the circuit and verify no short between 5 V and GND.
2. From the exact retained board revision's schematic and board inspection, draw the complete D13 factory-LED branch: PB7/D13, resistor value, LED polarity and electrical limits, and any buffer or other load between the MCU pin and indicator. Calculate its worst-case ON current from the measured rail and documented component limits. Include that branch in the per-pin, grouped-pin, and whole-device budget; "built in" does not mean load-free.
3. Calculate the worst-case D6 edge current from the identified transducer's capacitance/impedance, its primary source, the measured resistor, and the Mega output limits. Record the method and margin; the resistor alone is not an identity or current guarantee.
4. Attach the high-impedance waveform probe to TP-SND/D6 and its reference to Mega GND while unpowered. Connect USB, upload, and observe D13's 250 ms pulse and 750 ms dark separator before the cue.
5. At TP-SND/D6, record the cue's high/low levels, frequency, and finite duration. Interpret the waveform and audible cue together; hearing a sound alone does not establish frequency, duration, or pin state.
6. Wait for controlled shutdown. Host trace evidence must independently end with tone stop, D6 INPUT, D13 LOW, and D13 INPUT. Darkness and silence do not prove high impedance.
7. To prove D6's physical safe state, remove USB and install the measured 10 k Ω weak pull-up from the

measured 5 V rail to TP-SND/D6. Include probe/DMM input resistance, transducer leakage, and Mega leakage in the predicted released-voltage range. Reconnect USB, record TP-SND/D6 during the driven cue and after shutdown, then remove USB before removing the fixture. A released voltage in the calculated range is the physical high-impedance evidence.

8. Prove D13's physical post-shutdown state with a *separate*, board-specific weak-bias calculation. Using the identified factory branch, meter/probe impedance, MCU leakage limits, rail uncertainty, and component tolerances, choose and measure a temporary resistor from TP-ACQ/D13 to the rail that makes the predicted driven-LOW and released-INPUT voltage ranges non-overlapping. The fixture must remain within every D13 and factory-branch current limit even if firmware drives the opposite level. Install it only while USB is removed; reconnect USB, record D13 during the dark separator and after controlled shutdown, then remove USB before removing the fixture. If no safe resistor yields separable ranges for that board branch, do not claim physical high impedance; use a separately reviewed high-impedance current/impedance method instead.
9. With all fixtures removed, reconnect USB and press reset once. Confirm the acquisition pulse, separator, cue, and shutdown all repeat.
10. Disconnect USB at the computer end. Measure the 5 V rail, TP-ACQ/D13, and TP-SND/D6 relative to GND. Record the power-removed ranges before touching or dismantling the circuit.

Stop immediately and remove USB for an unidentified or warm part, unexpected continuous sound, a waveform outside the identified part or Mega limits, current above either calculated limit, overlapping weak-bias prediction ranges, a failed acquisition separator, or any failure to reach shutdown within 2 seconds after reset. Do not move a probe or change a weak-bias fixture while powered.

8.1 Safe fault exercises

Remove power before each wiring change, change one condition, then restore the exact schematic:

- disconnect D6: software reports sounding, but the transducer is silent;
- disconnect GND: same symptom, different open path;
- request 30 Hz: expect `InvalidArgument` and silence;
- pre-claim D6 or Timer2 in a host test: expect `ResourceBusy` before hardware work;
- call `shutdown()` during a long tone: expect immediate silence.

Do not short D6, omit the resistor as a “fault,” substitute a speaker, or probe the powered circuit with a meter in current mode.

9 Mega 2560 acceptance

Physical work remains pending. Leave every field blank until the named observation is made on the retained bench assembly.

Date/time: _____ Operator: _____

Reviewer/date: _____

Board maker/model/revision/serial or asset ID: _____

Port/FQBN: _____ ADK/sketch commit: _____

`arduino-cli`/AVR core versions: _____

Upload command: _____

USB source/cable/current limit and removal point: _____

DMM model/range/input resistance: _____

Scope/analyzer model, probe, attenuation, bandwidth, and reference: _____

Exact board schematic/revision and D13 branch topology: _____

D13 resistor/LED/buffer identities, values, tolerances, and sources: _____

Calculated D13 branch current/method/margin: _____
 Piezo maker/part/markings/passive-identity evidence: _____
 Piezo rated voltage and capacitance/impedance: _____
 Primary source and revision: _____
 100 Ω measured value/tolerance/rating: _____
 10 k Ω fixture measured value/tolerance/rating: _____
 Measured 5 V rail: _____
 Calculated worst-case D6 current/method/margin: _____
 Mega per-pin, grouped-pin, and total-device limits checked: ☐
 Flash: _____ bytes SRAM: _____ bytes

Evidence	Predict	Observe	Interpret
Acquisition	D13 pulse/separator; sounder silent		
Primary cue at TP-SND/D6	levels, 440 Hz, 250 ms		
Current/limits	D6 edge and D13 branch margins		
Controlled shut-down	tone stop; D6 and D13 re- leased		
Weak-pull D6 safe state	driven/released voltage ranges		
D13 safe state	board-specific bias ranges		
Reset	complete sequence repeats		
Power removal	5 V, D13, D6 unpowered ranges		

Unpowered continuity map and TP labels confirmed: ☐ Initialization status recorded independently: ☐
 Host trace records tone stop/D6 input and D13 LOW/input: ☐ D6 weak pull installed and removed only
 while unpowered: ☐
 D13 fixture resistor/value/rail: _____ opposite-drive worst-case current
 safe: ☐
 D13 driven-LOW/released predicted ranges: _____

D13 driven-LOW/released observed values: _____

D13 fixture installed and removed only while unpowered: ☐

Reset completed: ☐ USB physically removed before handling: ☐

Deviation and stop action: _____

Remaining uncertainty: _____

Result: ☐ draft ☐ rejected ☐ hardware verified Reviewer initials: _____

10 Explain and extend

Claim: Does the evidence support nonblocking, deadline-controlled sound?

Evidence: Cite a tone boundary, a concurrently updated component, and a safe shutdown observation.

Reasoning: Explain how explicit time and exclusive Timer2 ownership make the behavior predictable.

Limitation: What remains hardware experimental after host tests pass?

Next, map a few named cues to fixed frequencies without hiding **Status** results. Keep sequences deterministic: a cue table supplies data; the loop supplies time. Lesson 006 combines output, input, PWM color, and audible feedback in Simon.

11 References

The HTML lesson carries current commands, source links, API signatures, errata, and test navigation. This PDF emphasizes bench procedure and evidence.

- Arduino Mega 2560 Rev3 documentation and pinout
- Arduino tone melody example
- Arduino tone reference
- Microchip ATmega2560 datasheet
- ADK documentation